

One of the key battle areas is carried out at the mobile device operating level: Microsoft-lead Windows Mobile vs. Nokia-lead Symbian Series 60 systems. Nokia is currently in a strong position, but the true mass market for smartphones is only emerging. In the mobile game business, there is very strong competition originating also from the traditional game console manufacturers, such as Nintendo and Sony. These companies are adding communication features to their upcoming mobile game consoles (to be launched late 2004). This will surely intensify competition on the mobile game markets further. One additional dimension to the mobile battlefield is similar to the one seen in the early Internet years—the battle of the preferred mobile Internet browser. Norwegian Opera Software (supported by Nokia and Sony Ericsson) is taking a major challenge to compete against Microsoft's Mobile Internet Explorer (pre-installed on e.g. Orange Smartphones' SPVs). This competition will also have its impact on at least the mobile game sites browsing technology solutions. The current trend seems to be that WAP-based sites are becoming more and more XHTML-based sites. Multiple standards create major challenges for mobile game developers. A game created with one technology solution has to be re-created for another platform. Different screen sizes, varying sound solutions, operating system differences and memory shortages limit the possibilities for rapid game software transfer from one platform to another. Additional challenges can also be created by game crackers—the mobile device is no exception in its challenge of intellectual property right protection. These challenges influence directly the profitability of multi-platform game development. Several development companies have selected only a few technology solutions and game platforms they support.

One of the most interesting technology development areas in mobile gaming relate to multiplayer games. Mobile game devices can be connected to a gaming server. The server enables gamers to play against each other regardless of their geographical location. Nokia has created a full-scale mobile online game